Cambridge (CIE) A Level Business

Your notes

Operations Strategy (A Level)

Contents

- Operational Decisions
- Enterprise Resource Planning (ERP)
- Lean Production
- Operations Planning

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Your notes

Resources and operational decisions

- Operational decisions are impacted by, and have an impact on, business resources
- Business resources include:
  - Capital
  - Employees, including **contractors**
  - Property and land
  - Machinery and equipment, including IT hardware
  - Intellectual property such as **patents**
  - **Databases** and key software and hardware
- When operational decisions are made, care needs to be taken to **ensure that the appropriate resources are in place** or that adequate plans are made to ensure that they will be in place when needed
  - This may involve
    - Purchasing new facilities, such as land or property
    - Investing in new production equipment or hardware
    - Hiring new workers or training existing staff
    - Obtaining legal protection for inventions
    - Updating systems to ensure information is up-to-date and complete
- Without the appropriate resources, operations decisions may be **delayed** or require **significant amendment**
  - This may impact on the operations function's ability to achieve its overall targets and limit its contribution to achieving business **objectives**

Information technology and operational decisions

- Businesses are increasingly using a range of technologies to **automate production processes**

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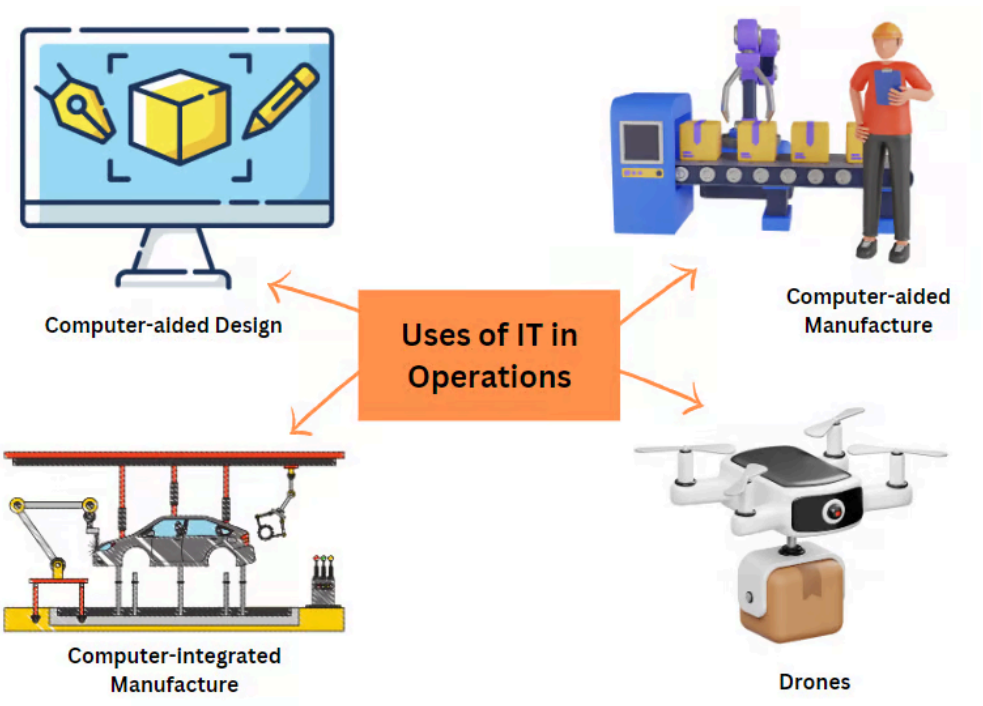

Information technology can be applied to a range of operations processes, including design, manufacture, safety and logistics

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1. Computer-aided design (CAD)

- The process of **digitally creating design simulations** of products in 2D or 3D
- It can also include the use of computer software to create **mapping processes** for industrial machinery, e.g. the movements required by a robotic arm in a television factory

Evaluating the use of CAD

Advantages	Disadvantages
▪ **Lower product design costs** as the need to produce expensive prototypes is eliminated ▪ **High levels of accuracy** reduce the chance of production errors and defects, leading to high **quality** output ▪ **Quicker design development** reduces the time taken to bring a product to market	▪ **Bespoke or complex design software** is usually very expensive ▪ Design and production employees need **extensive training** to ensure designs are transformed into finished products effectively ▪ Computer processing and storage capacity may need **upgrades**

2. Computer-aided manufacturing (CAM) and Computer-integrated manufacturing (CIM)

- **Computer-aided manufacturing (CAM)** is the use of computers to assist in operations of a manufacturing plant, such as **production** processes, planning, management, transportation and storage
- **Computer-integrated manufacturing (CIM)** is the **complete automation** of a manufacturing plant, with all processes functioning under computer control with digital information tying them together

Your notes

Evaluating the use of CAM and CIM

Advantages	Disadvantages
▪ Production can take place with **minimal human input**, reducing labour costs ▪ Often linked directly to CAD systems, it allows for **mass customisation** as changes are be made to work in progress ▪ IT-controlled production systems are usually **highly accurate**, generating high quality output	▪ High **initial cost** of hi-tech equipment that will need **ongoing maintenance** and upgrades ▪ **Equipment failure** and **breakdowns** can cause significant disruption to production ▪ **Output is only as good as the system's programmer** - if faulty code is not detected and corrected, product quality will be impacted

3. Drones

- Unpiloted aircraft are being used to carry out an increasing range of business tasks, including **package delivery, photography** and **security**
  - E.g. Large agriculture businesses use drones to spot failing crops and areas of concern in large fields, mapping boundaries and even applying pesticides

Evaluation of IT in operations

- The **installation and maintenance** of IT technology in production is likely to incur **significant costs**
  - Purchasing IT hardware and software may require external finance, such as **loans** which will need to be repaid with **interest**
  - Employees will need training in order to safely and effectively operate technology
  - **Down time** as technology is installed and workers are trained, it impacts output levels
- Businesses must be sure that this **capital expenditure is outweighed by gains in productivity** and **improvements in quality and flexibility**
  - Higher levels of productivity may only be achieved with **high levels of output**
  - Automation usually works best in the manufacture of **standardised products**

- Flexibility in production refers to how easy it is to install machinery and **switch to different machinery** which is required to manufacture different products (known as retooling)
    - The most flexible technology is also likely to be the most **costly**

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Artificial intelligence and operational decisions

- Artificial intelligence can be used in a range of ways to enhance business operations
    - In financial institutions such as **banks and insurance companies**, AI systems can identify customers' financial and behavioural patterns to identify potential instances of fraud
    - In **grocery retail**, AI systems can be used to monitor and control storage environments, such as freezers, to ensure that products are kept in optimum conditions
    - In **manufacturing**, AI can be used to schedule upgrades and maintenance to reduce the amount of production **down time**
- **Robots** are programmed to **collect information from their environment** using sensors and use artificial intelligence (AI) to improve production performance
    - E.g. Hotel chain **Yotel** employs 'robotic staff' that can move around anywhere in the hotels, carrying guests' luggage, delivering laundry, cleaning rooms and making coffee

Business uses of robots

Manufacturing

- In **manufacturing** and assembly lines, robots can increase productivity and accuracy

Logistics

- Warehouses and **logistics** businesses can use robots to move and transport goods

Agriculture

- In **agriculture,** tasks such as planting, harvesting and spraying crops can be carried out by robots

Medicine

- In **medicine** robots can be used in surgical procedures to increase precision and reduce the risk of errors

Entertainment

- Robots can be sold as toys or used in amusement parks to provide **entertainment**

Dangerous tasks

- Remote-controlled robots can be programmed to carry out **hazardous testing** and **R&D** processes

- The use of **robots** in particular presents a range of advantages and disadvantages related to productivity, quality and flexibility

Your notes

Evaluating the use of robots in operations

Advantages	Disadvantages
▪ Increased **productivity**, as robots can work continuously without breaks or fatigue ▪ Improved **consistency** and **accuracy**, as robots can perform tasks with a high level of precision and accuracy ▪ Robots can perform tasks that are **dangerous or hazardous** for humans, improving safety ▪ Robots can help **reduce labour costs** in the long run	▪ High **initial investment** of purchasing and setting up robotic systems ▪ **Maintenance costs**, as robots require regular maintenance and repairs, which can add to the overall cost ▪ **Lack of flexibility**, as robots are programmed for specific tasks and **cannot adapt easily** to new situations or tasks ▪ Increased use of robots can lead to **job losses** in some industries

The need for flexibility

- Businesses should have the flexibility to change output volumes, timescales and product specifications for several reasons
  - The level of **demand** can change with little warning
  - **Supplies** may become unavailable
  - Customers may **delay orders** or require **rapid delivery** of orders
  - Customers may demand **variations** to products
- Flexibility can be improved in a number of ways
  - **Outsourcing** some production to trusted manufacturers
  - Employing **multi-skilled workers** and investing in their training and development
  - Holding **buffer stock** to ensure increased demand can be met
  - Investing in **advanced machinery** that is capable of **mass customisation**
  - Investing in **buildings and land** to extend production premises

Examiner Tips and Tricks

> You may be asked to analyse the impact of IT and AI on a business. Their contribution to improving flexibility should be one of your key arguments - the case study may give you some ideas on how the business in question could apply them.

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Process innovation

- **Process innovation** involves identifying **new ways to organise and carry out production** processes
  - In recent years, organisations have made some notable improvements production processes that have improved productive efficiency

Examples of process innovation

Amazon	IKEA	UK Passport and Borders Agency
▪ **Amazon** provides detailed package tracking, allowing its online customers to provide details such as a safe place to leave deliveries, reducing the need to return undelivered items to warehouses	▪ **IKEA**'s **EPOS** system provides accurate stock availability information on its website, ensuring customers are not disappointed when they travel significant distances to purchase items	▪ In the public sector, the decision by the **UK Passport Agency** and **Borders Agency** to introduce biometric passports has allowed fast-track lanes at airports to be introduced, reducing queue times for travellers

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Features of an enterprise resource plan

- **Enterprise Resource Planning (ERP)** is a type of **integrated software system** that helps a business **manage and coordinate its core activities** — such as finance, production, supply chain, human resources, and customer relations — **all in one place**.
  - ERP solutions have evolved over the years, and many are now **web-based applications** that users can **access remotely**
- ERP can help a business **plan resources** by integrating:
  - planning
  - the purchasing of inventory
  - sales, marketing and finance
  - design and production and human resources
- It **collects information** about the activity and state of different production stages, allowing the different functional areas to **communicate and share information easily**

Common features of ERPs

Feature	Explanation
Integrated data management	▪ All departments share the same database, ensuring accurate and consistent data across the business
Modular design	▪ ERP systems have separate modules (e.g. finance, HR, inventory) that can be added or removed as needed
Real-time processing	▪ Data is updated instantly, helping managers make **faster and more informed decisions**
Automation of tasks	▪ Routine activities like invoicing or stock reordering are handled automatically, **saving time and reducing errors**
Customisation and scalability	▪ ERP systems can be tailored to business needs and scaled up as the business grows
Improved communication and collaboration	▪ All departments are connected, **improving coordination** and reducing misunderstandings or delays

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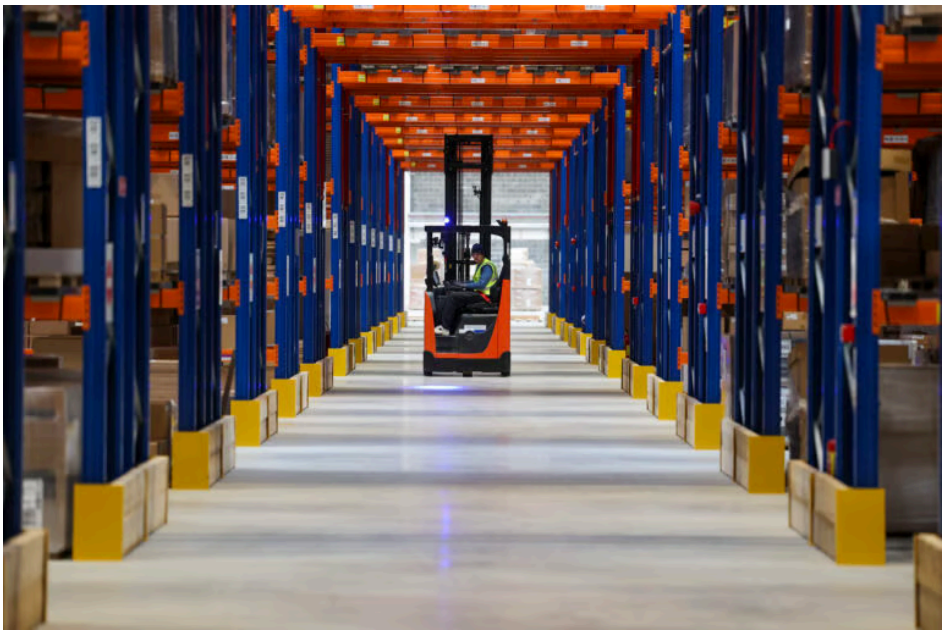
Case Study

IKEA's ERP system

- IKEA is a multinational furniture and homeware company based in Sweden, known for its flat-pack products and large warehouse-style stores



- With hundreds of stores in over 50 countries, managing operations efficiently is crucial
- Its ERP system has allowed IKEA to manage its **complex**, **global operations efficiently**, reducing costs and improving customer satisfaction

Why IKEA adopted ERP

- To improve **supply chain visibility** across its global operations
- To better manage **inventory levels** and avoid stock shortages or over-ordering
- To integrate various departments (finance, logistics, inventory) into **one unified system**
- To support **rapid international expansion** with consistent systems

The impact of ERP at IKEA

Area	Impact
Inventory management	▪ IKEA gained real-time insight into stock levels across stores and warehouses worldwide ▪ This helped **reduce waste** and improve product availability
Supply chain efficiency	▪ The ERP system helped coordinate deliveries from suppliers to stores more smoothly and cost-effectively
Financial control	▪ Improved accuracy in tracking costs, sales and profitability across its international markets

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Customer satisfaction	▪ More reliable stock availability and smoother logistics improved the overall shopping experience, increasing customer satisfaction

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ERPs and efficiency

- Enterprise Resource Planning (ERP) systems improve efficiency by connecting all key business functions, such as finance, operations, and HR, into one integrated system
- This allows for real-time data sharing, quicker decision-making and the more effective use of resources

How do ERPs generate efficiency?

1. Inventory control

- ERP systems monitor inventory levels in real time, across all warehouses and stores
  - This prevents overstocking and stock shortages, as stock levels are always accurate
  - It helps reduce waste and storage costs by only reordering when necessary

2. Costing and pricing

- ERP brings together data on purchasing, production and sales to give accurate cost information
  - This allows businesses to set realistic and competitive prices based on true costs
  - It also helps managers identify where costs are increasing and make changes to protect profit margins

3. Capacity utilisation

- ERP systems track machine use, labour hours, and output across departments or locations
  - This ensures that production facilities are neither underused nor overloaded
  - It helps businesses plan production more efficiently and make the most of their resources

4. Response to change

- Because ERP updates data instantly, businesses can react quickly to unexpected changes, such as supplier delays or a sudden rise in demand
  - This improves communication across departments and avoids disruption
  - It allows for faster and more confident decision-making

5. Workforce flexibility

- ERP systems store information on staff availability, qualifications and current roles
  - This makes it easier to move staff to different tasks or locations when needed

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- It also improves shift planning during busy or quiet periods

6. Management information

- ERP systems provide managers with accurate, real-time reports
  - This helps leaders monitor sales, costs, cash flow, and productivity without delay
  - It reduces time spent on manual reporting and improves the quality of decision-making

Limitations of ERP Systems

- Bespoke ERP systems are **expensive to install and maintain** and involve short-term upheaval to a business's operations.
- ERP systems **do not always eliminate inefficiencies** within a business and rarely improve every aspect of production
- A company's **reluctance to abandon old working processes** limits the effectiveness of ERP systems
- ERP systems cannot be successfully implemented during a period of **significant business change**

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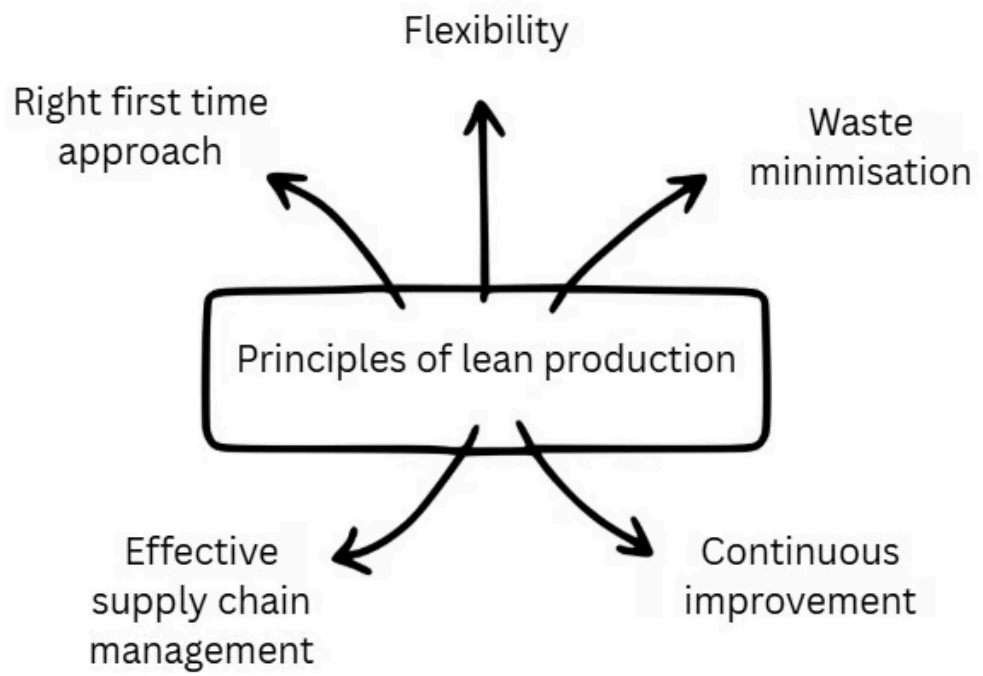
The aims and purposes of lean production

- Lean production is a management philosophy that aims to **maximise value** while **minimising waste**
- It focuses on **maximising efficiency**, **improving quality** and **reducing costs**
- Lean production is likely to lead to a competitive advantage
  - Lower **unit costs** are achieved due to minimal wastage so **prices may be lower** than those offered by competitors
  - **Better quality of output** is likely as a result of supplier reliability and carefully managed production processes

The main principles of lean production



Lean production involves adopting a right first time approach, flexibility, waste minimisation, effective supply chain management and continuous improvement

- **Right first time approach**
  - Aim for **zero defects** in output
  - Identify and solve problems as they arise
  - Prevent rather than correct errors
- **Flexibility**

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- Adaptable capital equipment and physical resources
- Multiskilled staff and team working
- Flexible management styles

- **Waste minimisation**
  - Remove processes that do not contribute to added value
  - Consume as little as is necessary
  - Rework rather than replace
- **Effective supply chain management**
  - Develop excellent relationships with suppliers
  - Minimal number of suppliers
- **Continuous improvement**
  - Ongoing, small steps
  - All staff involved in improvement

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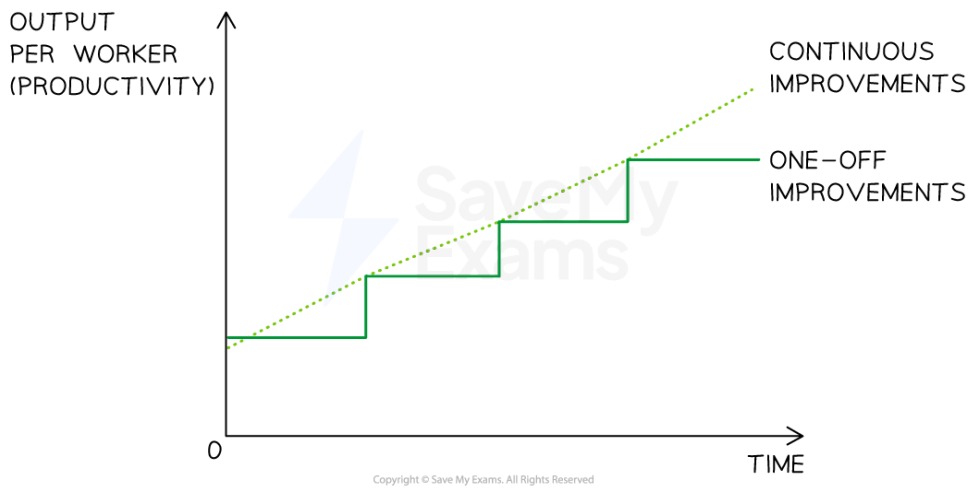
Lean production strategies

- Businesses that pursue lean production can adopt a range of strategies

1. Kaizen

- Kaizen means "continuous improvement" in Japanese
- It involves **small, regular changes** made by employees at all levels
  - Workers are encouraged to suggest ways to improve efficiency, quality and processes
  - These ideas are implemented gradually over time

A comparison of Kaizen with one-off improvements



Kaizen involves small, regular changes made by employees at all levels, implemented continuously over time

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Evaluating the use of Kaizen

Advantages	Disadvantages
▪ It involves all staff in improving the business ▪ It helps identify problems early ▪ It can boost employee morale and teamwork	▪ It can be time-consuming to collect and review employee suggestions ▪ Improvements may be small and take time to show results ▪ It requires a long-term commitment from management

2. Quality circles

- Small groups of employees who meet regularly to **discuss work-related problems** and **suggest improvements**
- These groups usually focus on improving quality, safety or productivity and include workers from different levels of the business

Evaluating the use of quality circles

Advantages	Disadvantages
▪ They encourage teamwork and employee involvement ▪ They generate practical ideas from people who do the job daily ▪ They can improve quality and reduce waste	▪ They may not be effective without strong leadership willing to implement suggestions ▪ They can be costly in time, training and time away from work ▪ Ideas may not always be implemented or taken seriously

3. Simultaneous engineering

- A production method where **product development stages happen at the same time**, rather than one after another
  - For example, design, engineering and marketing teams work together to reduce the time it takes to launch a new product

Evaluating the use of simultaneous engineering

Advantages	Disadvantages

▪ It can speed up product development ▪ Design issues can be identified early and rectified quickly ▪ It can provide a competitive edge with faster product launches	▪ It requires strong coordination between departments ▪ Mistakes in one area can affect others more quickly ▪ It can be difficult to manage in very large organisations

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4. Cell production

- A method of organising production into **small teams** that complete a whole part of the product process
- Instead of passing work along a production line, each cell is responsible for a complete unit of work, encouraging ownership and teamwork

Evaluating the use of cell production

Advantages	Disadvantages
▪ It can improve motivation through team responsibility ▪ It may also improve communication between workers ▪ It reduces movement of part-finished goods and waste	▪ The initial setup and employee training can be expensive ▪ It is not suitable for all types of production ▪ Team performance can be affected if members are absent

5. Just-in-time manufacturing

- Where **products are made only when there is demand**, and **each stage of production begins only when needed** by the next stage
- This approach aims to match production with customer demand
  - This minimises waste, speeds up workflows, and reduces the need for warehousing

Evaluating the use of just-in-time manufacturing

Advantages	Disadvantages
▪ It reduces waste by only producing what is needed ▪ It improves cash flow by avoiding overproduction and reduces cash tied up in stock ▪ It increases efficiency by streamlining workflows	▪ It is sensitive to disruptions in supply or production stages ▪ It requires very reliable suppliers and accurate forecasting of demand ▪ There is less flexibility to respond to sudden increases in demand

▪ It frees up space and reduces shrinkage, as goods are not stored between stages	▪ Delays at one stage can halt the entire production process

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6. Waste management

- Waste can be anything that does not add value, such as overproduction, excess inventory, delays or defects
- Lean waste management aims to simplify processes and reduce costs

Evaluating the use of waste management

Advantages	Disadvantages
▪ It can reduce costs as resources are used most efficiently ▪ It can help a business reduce its environmental impact ▪ It focuses on value-added activities, improving profitability	▪ It requires constant monitoring and careful evaluation ▪ Staff may resist changes or fear job losses when it is implemented ▪ It needs a strong culture of improvement to be effective

The impact of lean production

- Lean production affects not only how goods are made but also **how people work**, how **quality** is maintained and how **inventory**is managed

Lean production impacts

Area of Impact	Explanation

Inventory Control	▪ Lean production uses systems like Just-in-Time (JIT) to reduce the need for large inventories
	  ▪ This saves space, lowers storage costs and reduces waste
	  ▪ Businesses rely on accurate demand forecasting and reliable suppliers to avoid delays

Quality	▪ Lean production builds **quality into every stage,** not just at the end
	  ▪ Methods like Kaizen and quality circles help staff spot problems early
	  ▪ This results in consistent products and higher customer satisfaction

Employee roles	▪ Employees take an active role, suggesting improvements, working in teams and taking responsibility for quality ▪ This improves motivation but requires training and trust
Capacity management	▪ Lean production helps match output to demand, reducing waste and idle time ▪ Techniques like simultaneous engineering and cell production improve responsiveness ▪ However, responding to sudden demand increases can be challenging
Efficiency	▪ Lean production systems cut unnecessary steps and reduce delays using methods like flow production, JIT and waste reduction ▪ This leads to lower costs and faster production, and can allow a business to price products more competitively.

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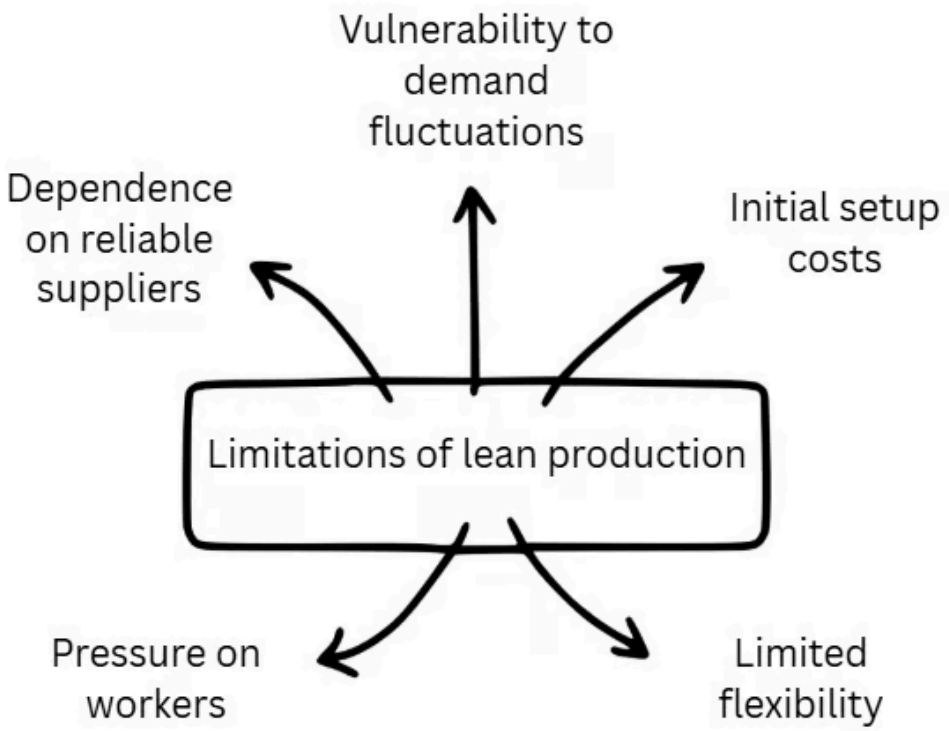
Limitations of lean production strategies

- Lean production can improve efficiency and reduce waste, but it also presents several **challenges and limitations** that businesses must carefully manage



Limitations of lean production include its initial setup costs and the pressure it can place on workers

- **Dependence on reliable suppliers**
  - Lean production often relies on just-in-time delivery
    - If suppliers are late or unreliable, production may stop entirely because there is little or no stock available
- **Vulnerability to demand fluctuations**
  - Lean systems produce only what is needed
    - This makes it difficult to respond quickly to unexpected increases in customer demand, which can result in missed sales and unhappy customers
- **Initial setup costs**
  - Introducing lean production may require investment in new equipment, staff training, and redesigning production processes
    - These upfront costs can be very high, especially for small businesses
- **Pressure on workers**
  - Lean production gives employees more responsibility and expects constant improvement
    - This can create stress or lower morale if staff feel overloaded or too closely monitored
- **Limited flexibility**
  - Lean strategies work best in high-volume, predictable environments
    - They are unlikely to be suitable for businesses with highly customised products or irregular demand

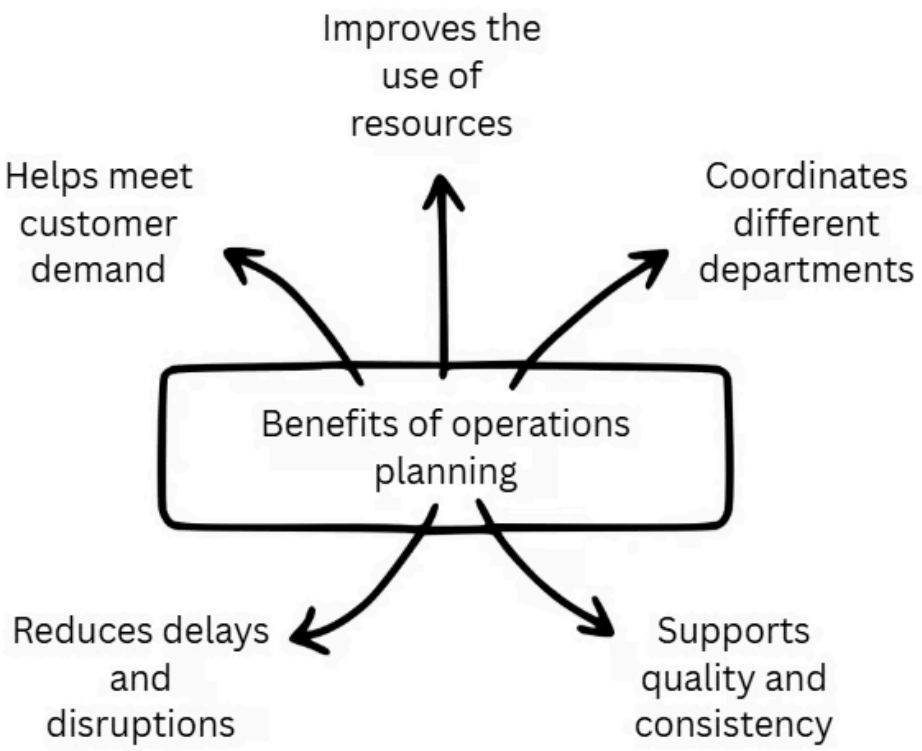
Your notes

Your notes

The need for operations planning

- **Operations planning** is the process of organising all the resources and activities needed to produce goods or services efficiently
  - It ensures that the right products are made, at the right time, in the right quantity and using the right resources

Benefits of operations planning



Operations planning helps a business meet customer demand and supports quality and consistency

- **Meet customer demand**
  - Planning helps businesses produce the correct quantity of goods to match expected demand
    - Without it, a business might make too little, causing stock shortages and lost sales,or too much, leading to waste and higher costs
- **Improve the use of resources**
  - Operations planning ensures that staff, machinery and materials are used efficiently
    - It helps avoid wasted time, overuse of equipment, or running out of stock during production

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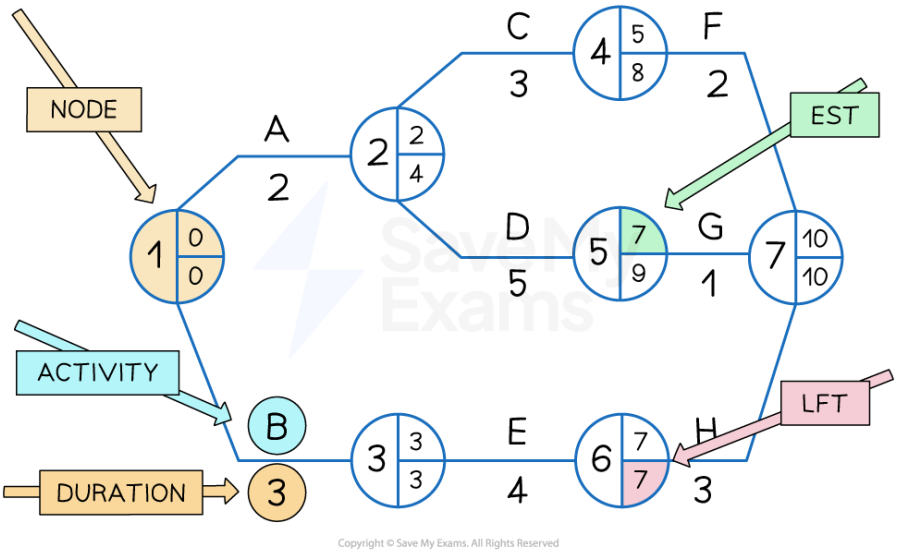
- **Coordination of departments**
  - Good planning allows production, purchasing, finance, and marketing to work together
    - E.g. Marketing can give sales forecasts, finance can set budgets and production can schedule output to meet targets
- **Reduces delays and disruptions**
  - Planning helps identify potential problems early, such as a shortage of raw materials or a machine breakdown
    - Solutions can be prepared in advance, reducing the risk of disruption
- **Supports quality and consistency**
  - Operations planning includes setting clear standards for quality, timing and workflows
    - This ensures that products or services are delivered consistently to meet customer expectations

Your notes

An introduction to network diagrams

- A **network diagram** is a visual planning tool used in operations management to help **organise and schedule tasks in a project**
- It shows the **order** in which activities must be completed and **how long** each one is expected to take

Example network analysis



A simple network diagram showing key elements

- A network diagram must **always start and end on a single node**
- **Lines must not cross** and must **only be assigned to activities**

How network diagrams help operations planning

- Network diagrams **improve time management** by helping managers identify the critical path, the shortest time needed to complete a project
  - They highlight **task dependencies** by showing which activities must be completed before others can begin, helping to avoid mistakes and delays
- Network diagrams support **efficient resource allocation** by showing when people, machines, or materials will be needed, reducing idle time and bottlenecks
  - They improve decision-making by making it **easier to adjust plans** if there are unexpected delays or changes to deadlines

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Elements of network diagrams

Node

- A **node** is a circle that represents a point in time where an activity either starts or finishes
- Each node is split into three sections
  - The **left half** of the circle shows the activity number
  - The **top right** section shows the **earliest start time (EST)**—the earliest point an activity can begin, based on the completion of the previous activity
  - The **bottom right** section shows the **latest finish time (LFT)**—the latest time by which the previous activity must be completed without delaying the project

Activities

- An **activity** is a task or process within a project that takes time to complete
  - Activities are shown on a network diagram as a **line linking two nodes**
  - Above the line, you will usually find either a letter or a short description of the activity

Duration

- The **duration** is the length of time required to complete an activity
- This is shown **below the activity line** and is usually measured in time units such as hours or days

Dummy activities

- A **dummy activity** is an activity that has a **weight** of **zero**
  - **Dummies** are **not** assigned names (letters)
  - **Dummies** are represented by **dotted lines**

Where a dummy activity is used

- To ensure each activity has a **unique pair** of **start** and **end nodes**

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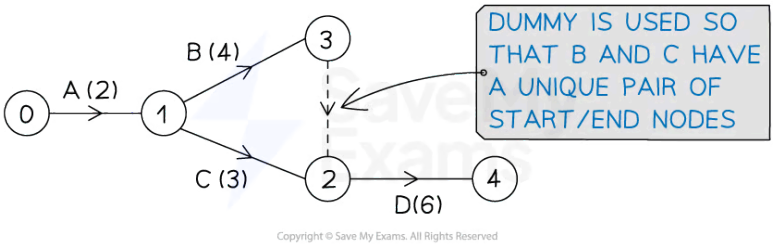
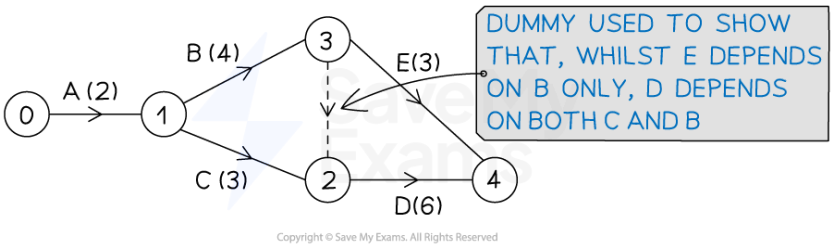
- E.g. In the activity network below, activity D has **immediate predecessors** B and C
  - B and C cannot **both** start at event (node) 1 and end at event (node) 2 (this would not be a unique pair)
  - a dummy activity is used so that B has start and end pair (1, 3) and C has a start and end pair (1, 2)



- When there is a split of **immediate predecessors**
  - E.g. In the activity network below, activity D has immediate predecessors B and C
    - Activity E only has B as an immediate predecessor
    - A dummy activity is used to show that D depends on both B and C



Using network diagrams for critical path analysis

1. Calculating earliest start times (EST)

- Working forward from Node 1, it is possible to calculate the **Earliest Start Time** for each activity by **adding the duration of each task**

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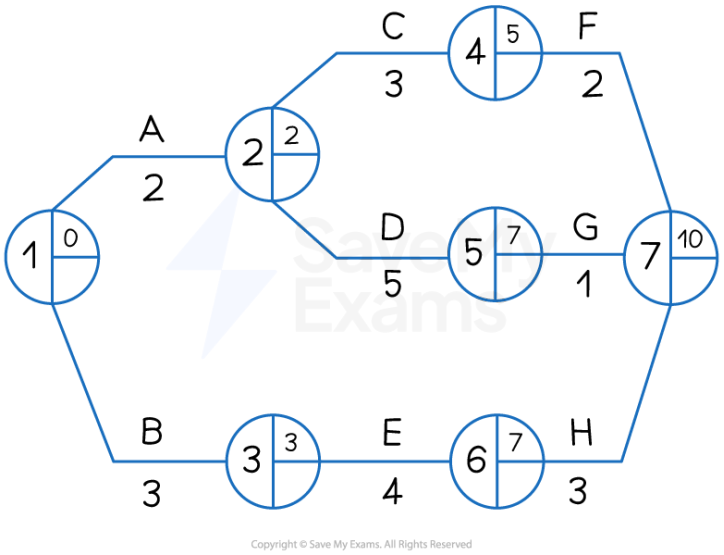


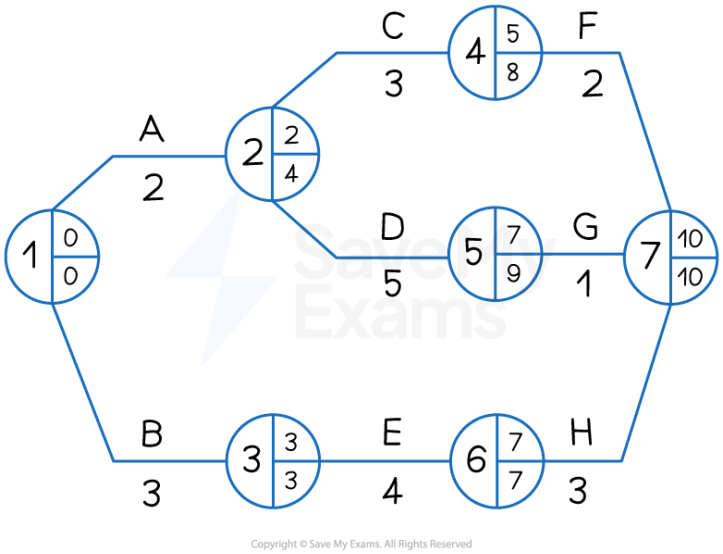
An example of a simple network diagram showing Earliest Start Times

- The EST for each activity is placed in the **top right** of each node
  - Node 1 is the starting point of the project and where both Activity A and Activity B begin
  - Activity A and Activity B are independent processes
  - Activity A has a duration of 2 days and its earliest start time (EST) is 0 days
  - Activity B has a duration of 3 days and its EST is also 0 days
  - Activity C and Activity D both begin at Node 2 and are dependent upon the completion of Activity A but are independent from each other
    - Activity C has a duration of 3 days and its EST is 2 days
    - Activity D has a duration of 5 days and its EST is also 2 days
- Activity E begins at Node 3
  - Activity E has a duration of 4 days and its EST is 3 days
- Activity F begins at Node 4
  - Activity F has a duration of 2 days and its EST is 5 days
- Activity G begins at Node 5
  - Activity G has a duration of 1 day and its EST is 7 days
- Activity H begins at Node 6
  - Activity H has a duration of 3 days and its EST is 7 days
- Node 7 is the end point of the project

2. Calculating latest finish times (LFT)

Your notes

- Working backwards from Node 7, it is now possible to calculate the Latest Finish Time (LFT) for each activity by **subtracting the duration of each task**



An example of a simple network diagram showing Earliest Start Times and Latest Finish Times

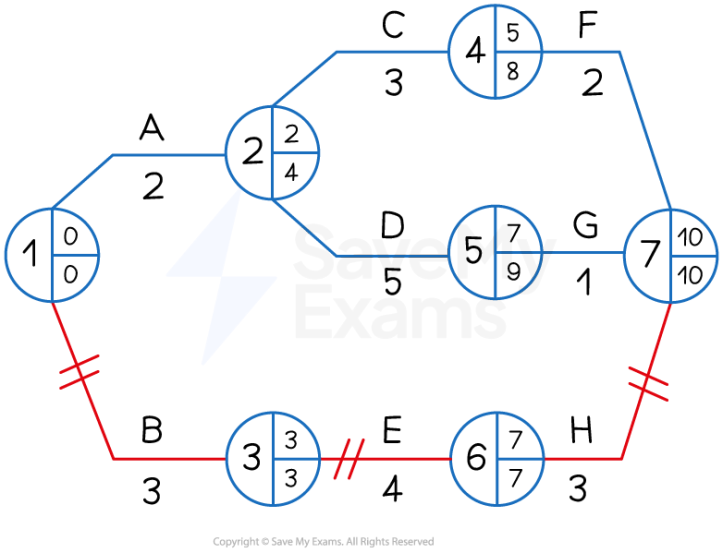
- The LFT for each activity is placed in the **bottom right** of each node
  - Node 7 is the end point of the project, which has a latest finish time of 10 days
  - Activity H has a duration of 3 days
    - The LFT in Node 6 is 7 days (10 days - 3 days)
  - Activity G has a duration of 1 day
    - The LFT in Node 5 is 9 days (10 days - 1 day)
  - Activity F has a duration of 2 days
    - The LFT in Node 4 is 8 days (10 days - 2 days)
  - Activity E has a duration of 4 days
    - The LFT in Node 3 is 3 days (7 days - 4 days)
  - Activity D has a duration of 5 days
    - The LFT in Node 2 is 4 days (9 days - 5 days)
  - Activity C has a duration of 3 days
    - The LFT in Node 3 is 4 days because Activity D is the more time-critical of the two activities that are dependent upon the completion of Activity A and so its LFT is recorded
  - Activity B has a duration of 3 days
    - The LFT in Node 1 is 0 days (3 days - 3 days)

Your notes

- Activity A has a duration of 2 days
  - The LFT in Node 1 is 0 days because Activity B is the more time-critical of the two starting activities and so its LFT is recorded
- The LFT in Node 1 is always 0

3. Identifying the critical path

- The **critical path** highlights those **activities that determine the length of the whole project**
- If any of these critical activities are delayed, the **project as a whole will be delayed**
- The critical path follows the **nodes where the EST and LFT are equal**
  - In the diagram below nodes 1 3 6 and 7 have equal ESTs and LFTs
  - Activities that determine these nodes are B E and H
  - These activities are marked with two short lines
  - The critical path is therefore BEH



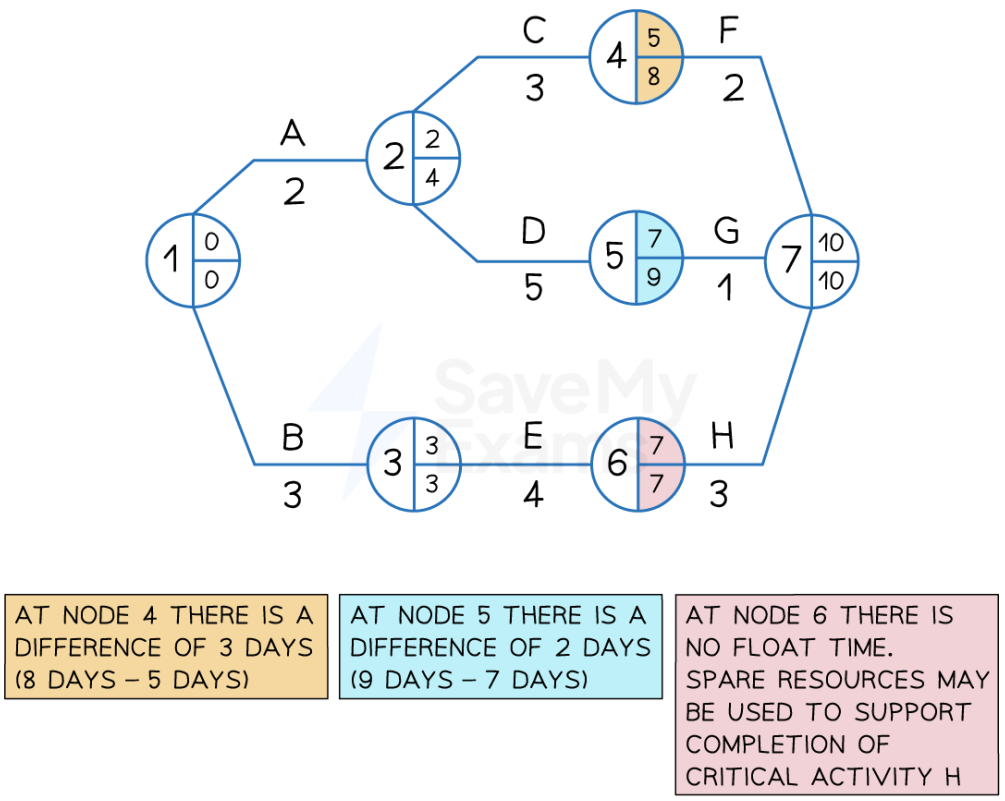
An example of a simple network diagram showing the critical path BEH

4. Identifying and calculating float time

- Float time exists where there is a **difference between the Earliest Start Time (EST** and the **Latest Finish Time (LFT)**
- Where float time is identified, managers may
  - **Transfer resources** such as staff or machinery to more critical activities
  - **Allow extra time** to complete tasks to improve quality or allow for creativity

Your notes

Your notes



An example of a simple network diagram showing float nodes (4 and 5) and a critical node (6)

Float time analysis

- The **total float** refers specifically to spare time that is available so that the **overall project completion is not delayed**
- The **total float** for a specific activity is calculated by

$$\text{Float} = \text{LFT for the activity} - \text{Duration of the activity} - \text{EST for the activity}$$

- Using the diagram above, the following **total float times** can be calculated for Activities A to H:

Activity	LFT	– Duration	– EST	= Total Float
A	4	2	0	2
B	3	3	0	0
C	8	3	2	3
D	9	5	2	2

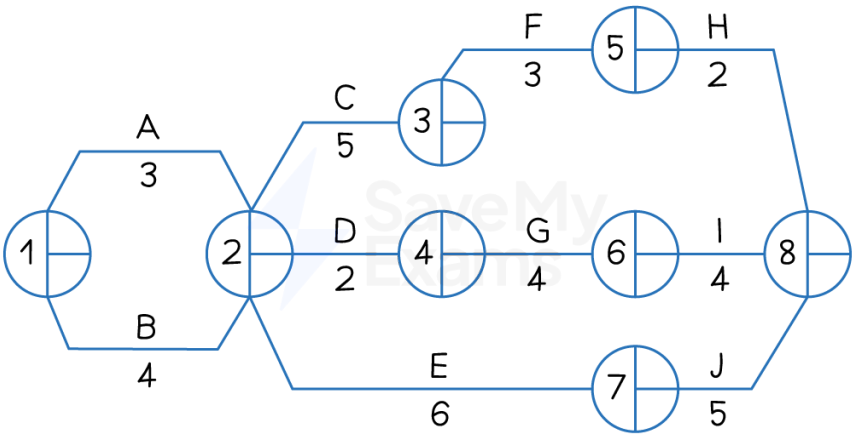
E	7	4	3	0
F	10	2	5	3
G	10	1	7	2
H	10	3	7	0

- The critical activities B E and H each have a total float of 0 days

Your notes

Worked Example

The network diagram below shows the activities involved in a new promotional campaign for a small fashion accessories business as well as the time (in weeks) it is expected that each activity will take to complete.



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Calculate

a) The earliest start times and latest finish times for each node.

[4]

b) The total float time for activity G

[2]

Step 1: Calculate the Earliest Start Times (EST)

Node 1 EST = **0**

Node 2 EST = 0 + 3 = 3 **but** 0 + 4 = 4 so **4**

Node 3 EST = 4 + 5 = **9**

Your notes

Node 4 EST = 4 + 2 = **6**

Node 5 EST = 9 + 3 = **12**

Node 6 EST = 6 + 4 = **10**

Node 7 EST = 4 + 6 = **10**

Node 8 EST = 12 + 2 = 14 **but** 10 + 4 = 14 **and** 10 + 5 = 15 so **15 (2)**

Step 2: Calculate the Latest Finish Times (LFT)

Node 8 = 15

Node 7 = 15 - 5 = 10

Node 6 = 15 - 4 = 11

Node 5 =15 - 2 = 13

Node 4 =11 - 4 = 7

Node 3 =13 - 3 = 10

Node 2 = 10 - 6 = 4

Node 1 = 4 - 4 = 0 **(2)**

Step 3: Calculate the total float time for Activity G

$$\text{Total float} = \text{LFT for the activity} - \text{Duration of the activity} - \text{EST for the activity}$$

$$= 11 \text{ weeks} - 4 \text{ weeks} - 6 \text{ weeks}$$

(2)

$$= 1 \text{ week}$$

Amending network diagrams

- If the length of time taken to complete an activity changes, there may be an effect on the critical path and available float

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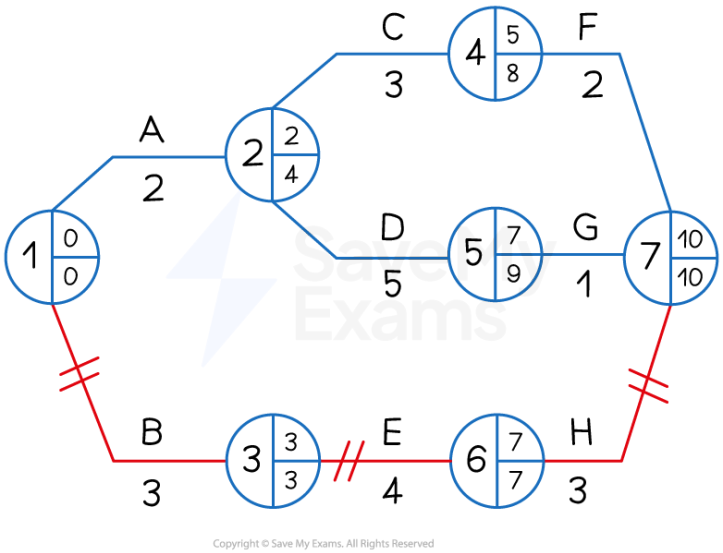
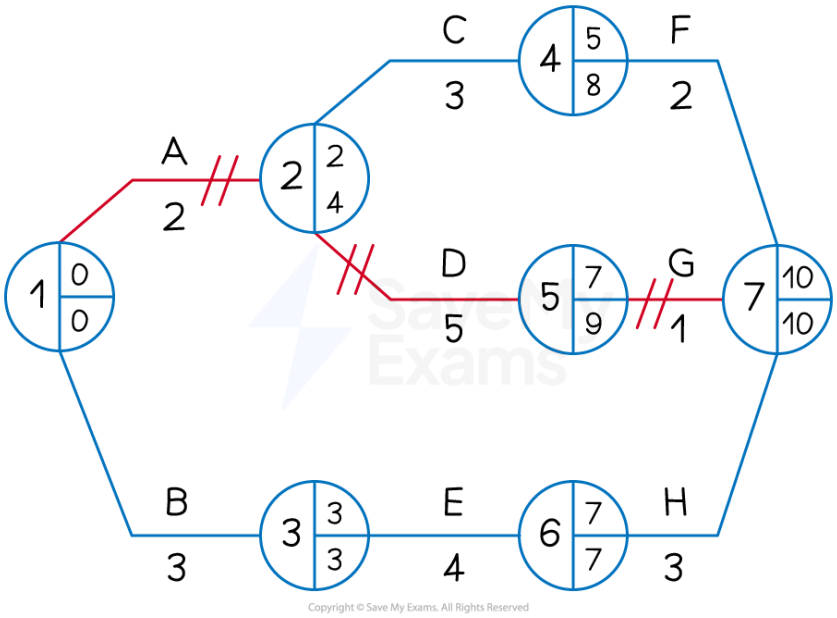


An example of a simple network diagram showing the critical path BEH

- In the strategy shown in the network diagram above, the duration of activity G is increased from 1 to 4 days
  - The EST at node 7 will increase to 11 days and the LFT to 11 days
  - The EST at node 5 will remains at 11 days, while the LFT will change to 7 days
  - The EST at node 5 will remains at 2 days, while the LFT will change to 2 days
- As a result, the critical path will switch from BEH to ADG



The amended network diagram showing the critical path ADG

Evaluating critical path analysis

Your notes

- Although critical path analysis can be useful in project planning, the method has some limitations

Your notes

Limitations of critical path analysis

Limitations	Explanation
▪ Very lengthy or **complex projects involve a very large number of activities** that have numerous dependencies ▪ Network analysis often **relies on estimates and forecasts** ▪ Network analysis **does not guarantee the success** of a project ▪ Resources may not prove to be as flexible as hoped when managers identify float periods	▪ **Supervisors and specialist network planning software** may be required ▪ Significant research and good communication with suppliers is required to make **a network diagram really useful** ▪ **Project managers** will need to be **highly skilled** and will need **experience** of working with complicated plans ▪ Employees may require **additional training** in order to transfer to critical tasks ▪ Machinery and other capital resources may need to be adapted or upgraded